

Nomad Digital

Connected Transport, Intelligent Solutions

Network Health Check Report

DOSTO Nahverkehr 6-car

Prepared for

OeBB - Oesterreichische Bundesbahnen

Trainset (Fzg. Nr.)	4736-108
Fahrzeug-ID (OeBB)	Fzg. 136
Configuration	6-car
CCU	box1-t8 (10.179.8.1)
Check date	2026-05-04
Author	Abbas Rizvi, Nomad Digital
Version	1.0
Status	Draft for review

Document Control

This document records findings from an on-train Layer-2 network health check performed on DOSTO trainset Fzg. 136 (4736-108) on 2026-05-04. Reviewers may add comments using Word Track Changes or the Reviewer Notes table in Appendix D.

Revision History

Version	Date	Author	Change Summary
1.0	2026-05-04	Abbas Rizvi, Nomad Digital	Initial draft - routine L2 health check on 4736-108 (Fzg. 136), including OBN config/firmware update and cabling fault assessment.

Approvals

Role	Name	Organisation	Date
Author	Abbas Rizvi, Nomad Digital	Nomad Digital	
Technical Reviewer		Nomad Digital	
OeBB Reviewer		OeBB	

1. Executive Summary

A Layer-2 network health check was performed on DOSTO trainset Fzg. 136 (4736-108) on 2026-05-04 by Abbas Rizvi (Nomad Digital). The check included switch inventory, RSTP topology verification, LLDP cabling audit, per-port error scan, OBN configuration and firmware update, and Westermo access point firmware update. Two cabling faults were identified via LLDP; one resolved by Stadler Rail, one outstanding.

Headline Verdict

OVERALL VERDICT: CABLING FAULT (one outstanding) - Layer-2 fabric operationally healthy via RSTP redundancy; full clean bill requires Stadler to reconnect the D1<->E2 inter-coach trunk cable.

- 18 VDS Rail Consist Switches found and reachable on management VLAN.
- Inter-coach trunks: e0-0 18/18 UP at 10G full-duplex. e0-1 17/18 UP - D1 e0-1 link-down (missing cable to E2, Stadler action required).
- RSTP root bridge: D1 (0/a0:59:3a:d0:5e:c0), 18/18 agreement - single stable root. RSTP routing around the D1<->E2 break; all ports FWD, no traffic loss.
- Per-port error scan: 1 port (C2 e0-1) with single-count noise (1x rx_err, 1x jabber, 1x runt, 1x giant - CRC=0, carrier=false=0). Not actionable.
- OBN: all 18 VDS switches on nv6-XX-v8-136 and firmware 7.4.2. F3 train_id corrected 144->136.
- Westermo APs: firmware 6.10.0-0 -> 6.11.2-0 update in progress (9/24 confirmed at time of report). APxm-v1 config for coaches 4-6 pending.
- Cabling fault 1 (C3 e0-0/e0-1 swap): RESOLVED by Stadler - LLDP confirmed correct.
- Cabling fault 2 (D1<->E2 missing link): OUTSTANDING - Stadler action required.

2. Scope and Methodology

2.1 What Was Checked

- All 18 VDS Rail Consist Switches on management VLAN 100 (10.179.8.128/25).
- All inter-coach trunk uplinks (e0-0 and e0-1 on each switch).
- LLDP topology check - e0-0 and e0-1 neighbour mapping vs. OBN expected topology.
- RSTP spanning tree root and port states across the fleet.
- OBN configuration and firmware audit (nv6-XX-v8-136 target, firmware 7.4.2).
- 24 Westermo AP firmware update (6.10.0-0 -> 6.11.2-0) and APxm-v1 config push for coaches 4-6.

2.2 What Is Out of Scope

- Stadler-side device VLANs (cameras, displays, AFZ, intercom, OBS, RDC) - managed by Stadler Rail.
- PWLAN / passenger Wi-Fi - separate Nomad Digital scope.
- Cellular WAN backhaul - separate Nomad Digital scope.
- Full L2 throughput sampling and Stadler FW reachability - deferred until D1<->E2 cabling resolved.

2.3 Method

The check was performed via SSH from the engineer laptop through the CCU (box1-t8, 10.179.8.1) to each consist switch. Switch CLI commands (show interface details, show lldp neighbours, show spanning-tree) were run per switch. OBN management used the OBN CLI on the CCU. All 7 known OBN software bugs were pre-patched via fix_obn.py before the update run.

3. Detailed Findings

3.1 Switch Inventory

Parameter	Observed	Expected
VDS switches found	18	18 (6-car consist)
Westermo APs found	24	24 (typical 6-car)
CCU vlan100 subnet	10.179.8.128/25	10.179.X.128/25
OBN train_id	136 (= 128 + 8)	Correct for this consist
OBN template family	nv6-v7	nv6-* (6-car)

3.2 Schema-to-IP Mapping

Switches identified by LLDP hostname (nv6-XX-v8-136) after OBN config push. Prior to push, identification was by trunk/access-port fingerprint.

Schema Role	IP	Notes
A1	10.179.8.185	
A2	10.179.8.194	
A3	10.179.8.182	Stadler FW switch - e1-4 multi-VLAN trunk
C1	10.179.8.190	
C2	10.179.8.192	
C3	10.179.8.180	Cable swap e0-0/e0-1 - now fixed by Stadler
D1	10.179.8.183	RSTP root; e0-1 link-down (cable missing to E2)
D2	10.179.8.195	
D3	10.179.8.186	
E1	10.179.8.179	
E2	10.179.8.191	e0-1 link-down (cable missing to D1)
E3	10.179.8.188	
F1	10.179.8.184	
F2	10.179.8.187	
F3	10.179.8.193	train_id corrected 144->136
B1	10.179.8.181	ZFR primary - e1-11 VLAN 2
B2	10.179.8.178	
B3	10.179.8.189	ZFR standby - e1-11 VLAN 2

3.3 RSTP Spanning Tree

Parameter	Value
Protocol	RSTP (Rapid Spanning Tree Protocol)
Root bridge	0/a0:59:3a:d0:5e:c0 (D1, 10.179.8.183)
Switches agreeing on root	18/18
Topology stable	Yes - single root, no TCN flapping observed
D1<->E2 trunk impact	E2 routes to root via E3->F2/F3->B-end (path cost 432100 vs normal 2000). All ports FWD. No traffic loss.

3.4 LLDP Cabling Audit

LLDP topology check run against OBN expected nv6-v7 topology. 36 trunk ports checked (e0-0 and e0-1 on all 18 switches).

Switch	Port	Live Neighbour	Expected	Status
C3 (10.179.8.180)	e0-0	A2	A2	RESOLVED
C3 (10.179.8.180)	e0-1	C2	C2	RESOLVED
D1 (10.179.8.183)	e0-1	NO NEIGHBOUR	E2	OUTSTANDING
E2 (10.179.8.191)	e0-1	NO NEIGHBOUR	D1	OUTSTANDING

All other 32 trunk ports: LLDP neighbours match OBN expected topology.

3.5 Inter-Coach Trunk Error Scan

All 36 inter-coach trunk ports checked. D1 e0-1 and E2 e0-1 are link-down (cable missing - no flapping, no errors). One noise event on C2 e0-1 (CRC=0, carrier=false=0 - not actionable).

Schema	Switch IP	Port	Speed	RX Errors	RX CRC	Jabber/Carrier-false
B2	10.179.8.178	e0-0	10000 Mb/s, Full	0	0	0 / 0
B2	10.179.8.178	e0-1	10000 Mb/s, Full	0	0	0 / 0
E1	10.179.8.179	e0-0	10000 Mb/s, Full	0	0	0 / 0
E1	10.179.8.179	e0-1	10000 Mb/s, Full	0	0	0 / 0
C3	10.179.8.180	e0-0	10000 Mb/s, Full	0	0	0 / 0
C3	10.179.8.180	e0-1	10000 Mb/s, Full	0	0	0 / 0
B1	10.179.8.181	e0-0	10000 Mb/s, Full	0	0	0 / 0
B1	10.179.8.181	e0-1	10000 Mb/s, Full	0	0	0 / 0
A3	10.179.8.182	e0-0	10000 Mb/s, Full	0	0	0 / 0
A3	10.179.8.182	e0-1	10000 Mb/s, Full	0	0	0 / 0
D1	10.179.8.183	e0-0	10000 Mb/s, Full	0	0	0 / 0
D1	10.179.8.183	e0-1	link DOWN (cable missing)	0	0	0
F1	10.179.8.184	e0-0	10000 Mb/s, Full	0	0	0 / 0
F1	10.179.8.184	e0-1	10000 Mb/s, Full	0	0	0 / 0
A1	10.179.8.185	e0-0	10000 Mb/s, Full	0	0	0 / 0
A1	10.179.8.185	e0-1	10000 Mb/s, Full	0	0	0 / 0
D3	10.179.8.186	e0-0	10000 Mb/s, Full	0	0	0 / 0
D3	10.179.8.186	e0-1	10000 Mb/s, Full	0	0	0 / 0
F2	10.179.8.187	e0-0	10000 Mb/s, Full	0	0	0 / 0
F2	10.179.8.187	e0-1	10000 Mb/s, Full	0	0	0 / 0
E3	10.179.8.188	e0-0	10000 Mb/s, Full	0	0	0 / 0
E3	10.179.8.188	e0-1	10000 Mb/s, Full	0	0	0 / 0
B3	10.179.8.189	e0-0	10000 Mb/s, Full	0	0	0 / 0
B3	10.179.8.189	e0-1	10000 Mb/s, Full	0	0	0 / 0
C1	10.179.8.190	e0-0	10000 Mb/s, Full	0	0	0 / 0
C1	10.179.8.190	e0-1	10000 Mb/s, Full	0	0	0 / 0
E2	10.179.8.191	e0-0	10000 Mb/s, Full	0	0	0 / 0
E2	10.179.8.191	e0-1	link DOWN (cable missing)	0	0	0
C2	10.179.8.192	e0-0	10000 Mb/s, Full	0	0	0 / 0
C2	10.179.8.192	e0-1	10000 Mb/s, Full	1 (noise)	0	1 / 0
F3	10.179.8.193	e0-0	10000 Mb/s, Full	0	0	0 / 0
F3	10.179.8.193	e0-1	10000 Mb/s, Full	0	0	0 / 0
A2	10.179.8.194	e0-0	10000 Mb/s, Full	0	0	0 / 0
A2	10.179.8.194	e0-1	10000 Mb/s, Full	0	0	0 / 0
D2	10.179.8.195	e0-0	10000 Mb/s, Full	0	0	0 / 0
D2	10.179.8.195	e0-1	10000 Mb/s, Full	0	0	0 / 0

3.6 OBN Configuration and Firmware

All 18 VDS switches updated to nv6-XX-v8-136 (config) and 7.4.2 (firmware). F3 template error corrected (train_id 144->136). All 7 OBN software bugs pre-patched via fix_obn.py.

Switch	IP	Config	Firmware
A1	10.179.8.185	nv6-A1-v8-136	7.4.2
A2	10.179.8.194	nv6-A2-v8-136	7.4.2
A3	10.179.8.182	nv6-A3-v8-136	7.4.2
C1	10.179.8.190	nv6-C1-v8-136	7.4.2
C2	10.179.8.192	nv6-C2-v8-136	7.4.2
C3	10.179.8.180	nv6-C3-v8-136	7.4.2
D1	10.179.8.183	nv6-D1-v8-136	7.4.2
D2	10.179.8.195	nv6-D2-v8-136	7.4.2
D3	10.179.8.186	nv6-D3-v8-136	7.4.2
E1	10.179.8.179	nv6-E1-v8-136	7.4.2
E2	10.179.8.191	nv6-E2-v8-136	7.4.2
E3	10.179.8.188	nv6-E3-v8-136	7.4.2
F1	10.179.8.184	nv6-F1-v8-136	7.4.2
F2	10.179.8.187	nv6-F2-v8-136	7.4.2
F3	10.179.8.193	nv6-F3-v8-136 (corrected from 144)	7.4.2
B1	10.179.8.181	nv6-B1-v8-136	7.4.2
B2	10.179.8.178	nv6-B2-v8-136	7.4.2
B3	10.179.8.189	nv6-B3-v8-136	7.4.2

3.7 Westermo AP Firmware Update

Firmware update 6.10.0-0 -> 6.11.2-0 initiated for all 24 APs. At time of report: 9/24 confirmed on 6.11.2-0. APxm-v1 config push to coaches 4-6 (12 APs) pending firmware completion.

Coach	AP	IP	Firmware Status
1	1	10.179.8.240	6.11.2-0
1	3	10.179.8.231	6.11.2-0
1	4	10.179.8.223	Update sent - completing
2	1	10.179.8.236	6.11.2-0
2	2	10.179.8.233	6.11.2-0
2	3	10.179.8.232	6.11.2-0
2	4	10.179.8.225	6.11.2-0
3	1	10.179.8.218	6.11.2-0
3	2	10.179.8.238	Update sent - completing
3	3	10.179.8.226	Update sent - completing
3	4	10.179.8.220	Update sent - completing
4	1	10.179.8.239	Update sent - completing
4	2	10.179.8.235	Update sent - completing
4	3	10.179.8.227	Update sent - completing
4	4	10.179.8.224	Update sent - completing
5	1	10.179.8.234	Update sent - completing
5	2	10.179.8.229	Update sent - completing
5	3	10.179.8.222	Update sent - completing
5	4	10.179.8.219	Update sent - APxm-v1 config pending
6	1	10.179.8.230	Update sent - APxm-v1 config pending
6	2	10.179.8.237	Update sent - APxm-v1 config pending
6	3	10.179.8.228	Update sent - APxm-v1 config pending
6	4	10.179.8.221	Update sent - APxm-v1 config pending

4. Risk Assessment and Recommendations

4.1 Risk Summary

Area	Risk Level	Rationale
Inter-coach Layer-2 fabric	LOW	All error counters clean. RSTP stable, single root.
D1<->E2 missing cable	MEDIUM	One inter-coach trunk broken. RSTP provides redundancy but resilience is reduced. Stadler action required.
AP firmware rollout	LOW	15/24 APs completing update at time of report. PWLAN may be degraded until complete.
Stadler-side beyond FW	UNKNOWN	Out of scope - Stadler Rail manages those VLANs.
PWLAN / cellular	NOT ASSESSED	Separate scope.

4.2 Recommendations

- Stadler Rail: reconnect cable between D1 port e0-1 and E2 port e0-1. After fix: run sudo obn discover && sudo obn validate on the CCU to confirm LLDP topology clean.
- Nomad Digital: once all 24 APs confirmed on 6.11.2-0, push APxm-v1 config to coaches 4-6 IPs.
- Once cabling resolved: perform full L2 health check (STP re-verify, error counters, Stadler FW reachability, throughput sampling).
- Use these findings as baseline for future checks. Repeat at 6-monthly intervals or after any maintenance.
- If end-users report packet loss, the Layer-2 fabric is confirmed healthy (D1<->E2 exception aside). Investigate PWLAN/cellular or Stadler-side.

Appendix A - Raw CLI Evidence

Verbatim CLI output for key ports.

C2 (10.179.8.192) - e0-1 noise event

```
Interface e0-1 is enabled, line protocol is up
Speed: 10000 Mb/s Duplex: Full
RX packets:90963984 TX packets:63434312
RX errors:1 runts:1 giants:1 frag:0 jabber:1
RX crc errors: 0 TX crc errors:0 carrier false:0
```

Note: 1 RX error against ~91M packets, CRC=0, carrier-false=0. Single corrupted frame at link init - noise, not actionable. C2 e0-1 connects to D3.

D1 (10.179.8.183) - e0-1 broken link

```
Interface e0-1 is enabled, line protocol is down
Speed: Auto Duplex: Auto
RX packets:0 TX packets:0
RX errors:0 runts:0 giants:0 jabber:0
RX crc errors: 0 carrier false:0
```

Note: Link simply down - no errors, no flapping. Cable physically disconnected.

C3 (10.179.8.180) - LLDP after cable swap fix

```
e0-0 a0:59:3a:d0:54:20 nv6-A2-v8-136 TTCMP bridge/switch
e0-1 a0:59:3a:d0:63:00 nv6-C2-v8-136 TTCMP bridge/switch
```

Note: e0-0 -> A2 (expected), e0-1 -> C2 (expected). Cabling fault resolved.

Appendix B - Glossary

Term	Definition
AFZ	Automatische Fahrgastzählung - automatic passenger counting
CCU	Communications Control Unit - Nomad Digital onboard server (Debian Linux)
DOSTO	Doppelstockwagen - Stadler Rail double-deck passenger train
Fzg.	Fahrzeug - German for vehicle/trainset
LLDP	Link Layer Discovery Protocol - L2 neighbour discovery
OBN	Onboard Network - Nomad Digital network management software on the CCU
RDC	Remote Diagnostic Centre - Stadler remote diagnostics
RSTP	Rapid Spanning Tree Protocol - L2 loop prevention and redundancy
VDS Rail	VDS Rail GmbH - manufacturer of consist switches on DOSTO trainsets
ZFR	Zugfunk Radio - onboard radio; B1/B3 e1-11 access port VLAN 2

Appendix C - Reference Documents

- OeBB IPv4 schema: ND-DEL-OB-035-IPA-136_NV_6Teiler.pdf (Fzg. 136, 4736-108)
- VDS Rail Consist Switch User Manual, version 2.0.4
- Nomad Digital DOSTO L2 Health Check Playbook (project CLAUDE.md)
- Nomad Digital fix_obn.py - OBN bug fix script (bugs 1-7)
- Stadler_4736-108_Cabling_Fault_Report_v1.0.docx - prior cabling fault report

Appendix D - Reviewer Notes

Use this table to log comments if not using Word Track Changes.

#	Date	Reviewer	Note / Action Required
1			

2			
3			
4			
5			